CommerceFlow Architecture Decisions

Version 1.0 | 29 October 2026 | Status: Design draft

ADR-001 Modular Monolith

Decision: Java Spring Boot modular monolith. Rationale: clear domain boundaries and simpler deployment and transactions than microservices. Consequence: enforce module boundaries.

ADR-002 PostgreSQL and Flyway

Decision: PostgreSQL 17 and Flyway migrations. Rationale: referential integrity and reproducibility. Consequence: migration validation with Testcontainers.

ADR-003 Session Authentication

Decision: Spring Security and Spring Session JDBC, HttpOnly session cookie and CSRF. Consequence: configure CSRF bootstrap, cookie flags, expiration, logout and proxy topology.

ADR-004 Product Inventory

Decision: stock quantity on products for MVP. Consequence: variants and warehouses out of scope.

ADR-005 Atomic Checkout

Decision: conditional stock decrement and order persistence in one transaction. Consequence: verify affected rows and rollback all writes on failure.

ADR-006 Idempotency

Decision: persistent idempotency keys scoped by customer, request fingerprint and status. Consequence: define concurrency, TTL and recovery. Infrastructure table is separate from 10 business tables.

ADR-007 Mock Payment

Decision: simulated payment outcome; no real charges or card collection. Failed mock payment makes no order or stock changes.

ADR-008 Automated Tests

Decision: risk-appropriate automated tests with required CI gates on feature PRs.

ADR-009 Branching

Decision: feature branches into develop, protected master reserved for validated releases.

Design artifacts are proposed specifications. No production implementation is asserted.

Approved Checkout Policy — ADR-006 Supplement

Retention

Completed checkout idempotency records must be retained for at least 24 hours after completion. After expiration, deduplication of old requests is not guaranteed.

Replay

A retry by the same authenticated customer with the same key and equivalent canonical request payload returns the stored result and must not create a second order.

Payload mismatch

Reusing a key with a different canonical request payload returns HTTP 409 Conflict without changing order or inventory state.

Concurrent request

Only one in-flight request per customer and idempotency key may execute checkout. Concurrent duplicates receive HTTP 409 Conflict with a retryable error response; the client should retry after the first request completes.

Cancellation

Only CONFIRMED orders may transition to CANCELLED. The cancellation and stock restoration execute atomically and stock must be restored exactly once.

Failure recovery

Idempotency state changes and order creation must be coordinated transactionally. Incomplete operations must have a defined recovery path and cannot be replayed as completed orders.

Mock payment

The mock payment service does not process real funds or collect card details. A failed mock payment creates no order and changes no inventory.